

Tejas R. Jammihal

Bioinformatics Scientist II, Children's Hospital of Philadelphia

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Bioinformatics scientist with seven years of experience developing and clinically translating computational methods, including three years in clinical genomics. Experience in cancer immunogenomics at Dana-Farber and MD Anderson includes co-first-authored publications in *Nature Cancer* and *Cancer Cell*.

EXPERIENCE

Bioinformatics Scientist II

Jun 2023 – present

Children's Hospital of Philadelphia, Division of Genomic Diagnostics, Translational Bioinformatics, Philadelphia, PA

- Develop and clinically translate methods for structural and copy-number variant detection in clinical genome sequencing, working with clinical geneticists and genetic counselors to improve diagnostic yield in rare pediatric disease.
- Built the bioinformatics function for CHOP's Undiagnosed Disease Program and co-designed its automated case workflow (data intake, DRAGEN processing, Emedgene prioritization) with the Arcus informatics team.
- Contribute to multi-omic and long-read diagnostic studies, including a genome–transcriptome re-analysis of Cornelia de Lange syndrome (npj Genomic Medicine 2026) and the WAFL long-read workflow (ACMG 2026 oral).

Associate Data Scientist

Jan 2022 – Jun 2023

MD Anderson Cancer Center, Houston, TX

- Led the computational analysis of exceptional responders to immune checkpoint inhibitors in renal cell carcinoma, integrating exome and RNA-seq data (Nature Cancer 2025, co-first author).
- Designed and validated pipelines to quantify human endogenous retrovirus expression from short- and long-read RNA-seq and applied them to lymphoma cohorts.
- Benchmarked long-read sequencing methods for isoform characterization, structural variant detection and antigen discovery.

Bioinformatics Analyst

Sep 2020 – Jan 2022

Dana-Farber Cancer Institute, Boston, MA

- Led the cross-trial validation of renal cell carcinoma molecular subtypes between two phase III trials using a random-forest classifier (Cancer Cell 2024, co-first author).
- Analyzed exome and RNA-seq data from more than 2,000 patients across clinical-trial cohorts and public datasets (TCGA, PCAWG).
- Containerized published pipelines and deployed them with WDL and Snakemake on GCP/FireCloud and AWS.

Computational Biology Co-op

Jun 2019 – Sep 2020

Novartis Institutes for BioMedical Research, Cambridge, MA

- Identified transcriptomic signatures of response to cigarette smoke in preclinical models and scored them in human disease data to inform indication expansion.
- Built an interactive Spotfire dashboard for discovery biologists and project managers.

EDUCATION

Master of Biotechnology

2018 – 2020

The Pennsylvania State University

M.Tech and B.Tech (dual degree), Biological Engineering, minor in Nanotechnology

2012 – 2017

Indian Institute of Technology Madras

PEER-REVIEWED PUBLICATIONS

* co-first author

1. Rajagopalan R, **Jammihal T**, Jadhav T, Kaur M, Szot K, Blair J, Raible SE, Conlin LK, Krantz I. Multi-omic re-analysis increases diagnostic yield in individuals with Cornelia de Lange syndrome. *npj Genomic Medicine* (2026). doi:10.1038/s41525-026-00602-y
2. Kilich G, Jadhav T, Maurer K, Breen C, **Jammihal T**, et al., Rajagopalan R, Sullivan KE. Novel biallelic LIG3 mutations causing lethal phenotype with immunodeficiency. *American Journal of Medical Genetics Part A* 200(7), 1662–1667 (2026). doi:10.1002/ajmg.a.70104
3. Josowitz R, Woyciechowski S, Jadhav T, **Jammihal T**, Linn RL, Rychik J, Gaynor JW, Rajagopalan R, Spinner NB. Placental malperfusion is associated with adverse outcomes in congenital heart disease and with genetic variants in placental developmental pathways. *JACC* 86(19), 1704–1720 (2025). doi:10.1016/j.jacc.2025.07.069
4. Kilich G, Maurer K, Jadhav T, **Jammihal T**, et al., Rajagopalan R, Sullivan KE. An unusual cause of hexokinase 1 deficiency: case report. *eJHaem* 6(4), e70123 (2025). doi:10.1002/jha2.70123
5. **Jammihal T***, Saliby RM*, Labaki C, et al., Van Allen EM, Choueiri TK, Braun DA, Shukla SA. Immunogenomic determinants of exceptional response to immune checkpoint inhibition in renal cell carcinoma. *Nature Cancer* 6(2), 372–384 (2025). doi:10.1038/s43018-024-00896-w
6. Saliby RM*, Labaki C, **Jammihal T***, et al., Shukla SA, Choueiri TK, Braun DA. Impact of renal cell carcinoma molecular subtypes on immunotherapy and targeted therapy outcomes. *Cancer Cell* 42(5), 732–735 (2024). doi:10.1016/j.ccell.2024.03.002
7. Barroso-Sousa R, Forman J, Collier KA, Weber ZT, **Jammihal TR**, Kao KZ, et al., Shukla SA, Tolaney SM. Multidimensional molecular profiling of metastatic triple-negative breast cancer and immune checkpoint inhibitor benefit. *JCO Precision Oncology* 6, e2100413 (2022). doi:10.1200/PO.21.00413

PREPRINTS

1. Calame DG, Wiener EK, et al., **Jammihal T**, et al., Lupski JR, Gibbs RA, Rajagopalan R, Vanderver A. PTPN1-related autoinflammation is a common cause of Aicardi-Goutières syndrome with reduced penetrance. *medRxiv* (2026). doi:10.64898/2026.03.27.26345228

SELECTED PRESENTATIONS

- WAFL: a comprehensive, multi-omic computational workflow for rare disease diagnostics using long read sequencing. ACMG Annual Clinical Genetics Meeting (2026). *Oral presentation*.
- Determinants of exceptional response to immune checkpoint inhibition in metastatic clear cell renal cell carcinoma. ESMO Congress (2023). *Best Poster, Renal Cancer (Kidney Cancer Association)*.
- Clonal neoantigen load and tertiary lymphoid structure formation are associated with exceptional response to immune checkpoint inhibition in clear cell renal cell carcinoma. Kidney Cancer Research Summit, Philadelphia (2022). *Selected oral presentation*.

Full list of 12 conference abstracts at tejasrj.io/publications.

SKILLS

Data	WGS, WES, bulk and long-read RNA-seq, scRNA-seq, spatial transcriptomics, ATAC-seq
Variant analysis	DRAGEN, GATK, MuTect2, Manta, CNVPytor, ABSOLUTE, FACETS, mutational signatures, Emedgene, IGV
Immunogenomics	HLA typing (POLYSOLVER), neoantigen prediction (NetMHCpan), TLS signatures, immune deconvolution (CIBERSORTx)
Statistics and ML	survival analysis, multivariable regression, random forest, CNN transfer learning, PyTorch, AutoGluon
Programming	Python, R, Bash, SQL, JavaScript
Infrastructure	WDL, Snakemake, Nextflow, Docker, Singularity, AWS, GCP/Terra, SLURM, LSF, Git
AI-assisted development	Claude Code and the Claude API, with generated code reviewed and validated before use

HONORS

- Best Poster, Renal Cancer category, Kidney Cancer Association, ESMO 2023
- Oral presentation, ACMG Annual Clinical Genetics Meeting 2026
- Selected oral presentation, Kidney Cancer Research Summit 2022

SERVICE AND MENTORING

- Invited peer reviewer: Nature Biotechnology, JCO Clinical Cancer Informatics, Scientific Reports, Therapeutic Advances in Medical Oncology
- Peer mentor for new hires in immunogenomics, MD Anderson Cancer Center (2022 – 2023)